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Examiner_Final_report.pdf

File Size

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Question 1

Of the handful of results that did come back significant such as the hardness accuracy gap, the dissolution improvement from sharing one encoder, the attribution shift on dissolution and hardness; how confident can we be that these are real, given how many tests were run without any correction for that?

Answer 1:

You are right, and I should have applied a correction. I did not, and that was an oversight rather than a decision. The report contains no mention of Benjamini–Hochberg, Bonferroni or any other adjustment.

Twelve significance tests were run in total: four in RQ1 comparing MT-TrajNet against tuned gradient boosting, four in RQ2 comparing multi-task against single-task, and four in RQ3b for the attribution rankings, one per target. I have now corrected across all twelve.

Test	Raw p	Benjamini–Hochberg	Bonferroni
Attribution, dissolution	0.0004	0.0024 survives	0.0048 survives
Attribution, hardness	0.0004	0.0024 survives	0.0048 survives
Multi-task gain, dissolution	0.012	0.045 survives	0.144 fails
Hardness gap vs gradient boosting	0.015	0.045 survives	0.180 fails
Attribution, weight RSD and tensile	0.073	0.145 fails	0.871 fails

Here is where I stand on each of the three you name.

The attribution p -value survives both corrections comfortably. But I should be careful about what that actually means, because it is easy to read more into it than it can carry. What the test shows is that the two attribution rankings, confident and unconfident, are related rather than independent. The claim the report built on top of it, that the model leans much harder on particular channels when it is unsure, is a separate thing and never had a significance test of its own. So the surviving p -value supports less than the report implied. I go into that properly under Question 8.

The dissolution multi-task gain and the hardness accuracy gap both come out at 0.045 under Benjamini–Hochberg, just inside the boundary, and both fail Bonferroni. They are borderline. I should have described them that way in the report rather than calling them significant without qualification.

One detail about the count, since it changes the arithmetic. In RQ3b, dissolution and hardness return the same statistic, and weight RSD and tensile return the same as each other. That is a coincidence of sample size, not a duplicated test. With ten channels, Kendall's τ is a function of the discordant-pair count, so it can only take 46 distinct values: $\tau = 0.822$ means exactly 4 discordant pairs, and $\tau = 0.467$ means exactly 12. I checked this rather than assuming it. If those pairs are instead counted as two tests rather than four, giving ten in total, the same findings survive each correction, so the conclusion does not depend on how they are counted.

One thing in my favour, though it does not cancel out the mistake. The report already treats every non-significant result as inconclusive rather than as proof that two models are equal,

because three folds give the test so little power. The correction makes me more careful, but it does not reverse anything.

Question 2

Your work treat product code 15 as the key test of how the model handles a genuine distribution shift (Section 3.5, Section 6.3) and finds the error jumps to 2.43 times the baseline on it. But code 15 was also the code entirely excluded from training in the first place, specifically because it was too large and would dominate a fold (Section 3.4). So, when the model does badly on code 15, it's not clear whether that's because the model genuinely struggles with an unfamiliar hardness regime, or simply because it never saw a single example anything like it during training which I think is a much less interesting and less surprising finding. How would this "distribution shift" result look different from ordinary "the model has literally never seen this" behaviour, and is there a way to tell those two explanations apart?

Answer 2:

This is the confound the experiment was designed around, and code 25 exists specifically to tell the two explanations apart.

Both codes are unseen by the fold 1 model. Neither appears anywhere in its training data. The only thing that differs between them is where their hardness sits.

	Code 25	Code 15
Seen anywhere in training	No	No
Batches	34	64
Mean hardness	50.4 N	90.2 N
Hardness RMSE	4.918 N	18.69 N
Against CV reference of 7.698 N	0.64×	2.43×

That table is how you tell the two apart. If novelty alone were the explanation, code 25 should be penalised as well, because it is exactly as novel. It is not. Its error is about a third below the cross-validation baseline. So an unfamiliar product on its own costs nothing here. Novelty is held constant across the pair, only the hardness regime differs, and only the regime shift produces a penalty.

There is stronger evidence than a single pair. Because the folds are grouped by product code, all 24 codes inside cross-validation are themselves leave-one-code-out results at no extra cost, and every one of them was scored by a model that had never trained on it. If novelty were the driver, all 24 should be penalised.

Grouping all 24 by distance from the centre of the training hardness distribution, the pattern runs in one direction across the whole range rather than only at the ends:

Distance from training centre	Codes	Mean hardness RMSE
Within 3 N	12	4.55 N
Between 3 N and 10 N	5	5.38 N
More than 10 N	7	24.65 N

Across all 24, error against distance from the centre gives $r = 0.941$ with $p = 7.7 \times 10^{-12}$. That is Figure 3b in the report.

Since some of those codes are small, I checked whether the correlation depends on them. Seven of the 24 have five batches or fewer, together holding 18 of the 941 batches. Weighting each code by its batch count gives $r = 0.856$, and restricting to the 10 codes with at least 20 batches gives $r = 0.947$. So the relationship is not an artefact of a few tiny codes. I should add that the relationship is not linear across the whole range, since the error climbs much faster at the far end than near the centre, which is what the three bands above show.

All 24 codes are equally novel, so novelty cannot explain a correlation that strong. Distance from the trained range can.

One thing I will concede. Code 15 sits at the far end of that line and it was held out for a structural reason, so on its own it is not a clean test, and you are right about that. The 24-code analysis is the stronger evidence and I should have led Section 6.3 with it rather than presenting code 15 as the headline result.

Question 3

Why did one weak result (hardness accuracy) get a full root-cause investigation while the other one (weight uniformity's uncertainty calibration - roughly five times worse than the other three targets) was just noted and left as a limitation? Was there an attempt to understand why weight uniformity's intervals don't respond to the requested confidence level, or was that simply out of scope?

Answer 3:

It was out of scope, and honestly that came down to time rather than a proper decision. Hardness appears in the main accuracy table, so a reader meets it first, and that is where I spent the effort. The weight uniformity calibration problem is one line in a table, and I flagged it without chasing it. There was no attempt to find the cause before submission.

I have looked at it since. The behaviour is exactly as you describe. Weight RSD coverage moves only from 0.930 at the 80 per cent nominal level to 0.960 at 95 per cent, a spread of 0.030, against 0.127 for dissolution, 0.111 for hardness and 0.075 for tensile. It is already over-covered at 80 per cent and barely responds after that.

I have a hypothesis about the cause, which I want to label as a hypothesis before I give it, because I have not tested it. It concerns the shape of the target. Weight RSD is the only heavily skewed one of the four. Its median is 0.97, its 95th percentile is 1.86, and its maximum is 9.90, so the maximum is more than ten times the median. The other three targets have maximum-to-median ratios between 1.1 and 1.9.

The evidential head assumes a Normal-Inverse-Gamma predictive distribution, which is symmetric. Fitting a symmetric interval to a skewed target means one multiplicative scale factor cannot serve both the dense centre and the long tail. The scale large enough to cover the tail over-covers everything else, which is what a flat coverage curve looks like.

To be clear about what that paragraph is: a hypothesis that fits the evidence, and nothing more. It is not a finding, and I did not test it. The experiment that would confirm or refute it is to model log weight RSD instead and check whether the coverage curve straightens out. I have not run that experiment, so I cannot tell you whether the explanation is right.

The part I am sure about is what the report already says. Those intervals should not be relied on for an individual batch. The uncertainty-error correlation for weight RSD is +0.029, essentially

zero, so the interval carries no information about which particular batch is likely to be wrong.

Question 4

You evaluate your model against a deep ensemble baseline to assess the quality of its uncertainty estimates. However, unlike the rest of the study, this comparison is restricted to a single fold rather than the full three-fold cross-validation. While you acknowledge that this provides only a preliminary indication, the conclusion still asserts that the proposed model captures “most of the benefit” of the ensemble at significantly lower computational cost.

Answer 4:

You have identified a real inconsistency in the report, and I accept it.

The sentence in Section 6.3 reads that the single-pass evidential model “provides most of the benefit at roughly one-fifth of the training and inference cost”. There are two claims in that sentence, and they are not equally well supported.

The cost claim is just arithmetic, so it holds no matter how many folds were run. A five-seed ensemble is five training runs and five forward passes against one, so one-fifth is a count, not a measurement.

The benefit claim is a different matter. “Most of the benefit” is quantitative, and a single fold cannot support a quantitative claim. Labelling the comparison as directional in the following sentence does not repair it, because the strong claim has already been made. I should have written that the comparison is preliminary and indicates a direction, without attaching a size to it.

The underlying numbers are also more mixed than that phrase suggests. Under distribution shift on code 15, the ensemble is more accurate on all four targets:

Target	Ensemble	Evidential	Ensemble better by
Dissolution	2.793	2.895	3.5%
Hardness	14.08	18.69	24.7%
Weight RSD	0.342	0.364	6.0%
Tensile	0.117	0.143	18.2%

In distribution on the same fold, however, it reverses on two targets. The evidential model gives 3.344 on dissolution against the ensemble’s 3.408, and 7.900 on hardness against 9.081.

So the corrected statement is this. On one fold, the ensemble is more robust under distribution shift on all four targets, and clearly so on hardness. The cost ratio stands. The phrase “most of the benefit” puts a number on something measured once, and it should not have been written that way.

Question 5

Given the single-fold evaluation, how robust is this conclusion? Why was the full three-fold evaluation not conducted for this baseline to ensure parity with the primary results?

Answer 5:

Taking the two parts separately.

How robust the conclusion is. Not robust enough for how strongly the report put it. There is a single observation per target, so there is no spread to inspect and no test to run. The direction is at least consistent, since the ensemble is more accurate on all four targets under shift on code 15, rather than winning on some and losing on others. But four targets from the same fold are not four independent observations, since they share the same model, the same held-out data and the same seed.

Table 4 in the report shows why one fold is particularly weak here. MT-TrajNet's own fold-to-fold spread is $1.06\times$ on dissolution and hardness, but $1.74\times$ on weight RSD and $1.43\times$ on tensile. On the two noisier targets, the variation between folds is of the same order as the differences I am reporting between models. A single fold cannot separate a real effect from that variation.

The one part I would defend is the hardness comparison, 14.08 N against 18.69 N, a gap of 24.7 per cent. That is larger than MT-TrajNet's own $1.06\times$ hardness spread across folds, so it is unlikely to be pure fold noise. The other three gaps, at 3.5, 6.0 and 18.2 per cent, I would not defend on one fold.

Why three folds were not run. Compute. A five-seed ensemble means five full training runs per fold, so three folds would have required fifteen runs against the three the main model needed. The work ran on the free Kaggle tier with a twelve-hour session limit and a single Tesla T4. Fifteen runs alongside the rest of the study was more than I could complete reliably, so I ran fold 1 and labelled the result as directional.

I should add two things that make that explanation less comfortable than it first sounds. The five seeds came from my proposal, which specified a five-seed ensemble, so the number was carried over rather than chosen for this experiment. And fifteen runs was not the only alternative to one fold. Three seeds across three folds would have been nine runs, fewer than fifteen, and would have given me fold coverage at the cost of a smaller ensemble. I did not consider that trade-off at the time, and there is no record of it anywhere in my notebooks. Had I thought about it, fold coverage would probably have been the better use of the compute, because the fold-to-fold variation is exactly what the single-fold result cannot speak to.

I still think leaving it at one fold was a fair call given the compute I had. What was not reasonable was letting a single-fold result support a conclusion phrased as strongly as it was. If I were to do it again with the same compute, I would either run the ensemble on all three folds and drop something else, or keep it on one fold and state only the direction.

Question 6

The hardness root cause analysis in Section 6.5 explains the model's weak hardness accuracy by pointing to how few high hardness batches remain once code 15 is withheld. But that same scarce, code concentrated hardness data is also what produces your strongest positive result, the attribution shift with tau equal to 0.822 on hardness in Section 6.4. Is it possible that the same structural quirk, a handful of codes driving both the high hardness examples and the uncertainty signal, is inflating the attribution finding, rather than that finding reflecting genuine learning of physically relevant sensors?

Answer 6:

This is the hardest question of the ten, and when I went and checked, part of what you are pointing at is true.

The fold 1 test set has 314 batches. Code 4 is the one high-hardness code in it, with 22 batches. Splitting those 314 batches into quintiles by hardness uncertainty, **all 22 code 4 batches fall in the least-confident fifth and none fall in the most-confident fifth.** Code 4 therefore

makes up 35 per cent of the low-confidence group and 0 per cent of the high-confidence group. The concentration you describe is real and it is present in my data. I had not checked this before submission and I should have.

But it is not the whole story, and the test that separates the explanations is available. If the finding were purely a code 4 artefact, removing code 4 should destroy it.

Excluding code 4 and recomputing the quintiles on the remaining 292 batches, both groups become dominated by the same product code, code 13, at 86 per cent of the low-confidence group and 81 per cent of the high-confidence group. Their mean true hardness becomes 50.95 N and 51.14 N, essentially identical. Yet the model still separates them by a factor of 3.34 in its own uncertainty, 41.14 against 12.30.

The cleanest version of the test stays inside one product code. Code 13 has 131 batches, all the same product, all in the low-hardness range:

Within code 13 only	Least confident 20%	Most confident 20%
Mean uncertainty	51.40	10.69
Mean true hardness	50.4 N	52.8 N
Mean absolute error	9.12 N	4.38 N

Same product, same hardness range, and the model still separates confident from unconfident batches by 4.8 times in uncertainty. More importantly, that separation tracks reality: the batches it is less sure about are the ones it actually gets more wrong, by a factor of 2.08.

So the uncertainty is not just another way of saying “this batch comes from a high-hardness code”. It operates within a single product code, at a single hardness level, and it is informative about error there.

There is a limit to how far I can take this, and I would rather say so than leave it unsaid. I cannot rerun the SHAP attribution itself with code 4 excluded, because I saved the aggregated per-channel ratios rather than the per-batch SHAP values. So I can show that the underlying uncertainty signal survives the confound, but not directly that the specific ratios of 4.86 and 2.81 survive it. Re-running that analysis with code 4 excluded would be the definitive test, and it is the first thing I would do if extending this work.

Question 7

Your calibration correction is cross fitted so that no batch is calibrated on its own prediction (Section 3.6), and you note that an earlier, leakier version produced almost the same coverage numbers. Does that similarity not raise a concern about how sensitive this calibration step actually is? If the correction were doing meaningful work, would you not expect the numbers to move more once the leak was fixed? What would tell you the calibration step is not effectively a no op given your sample size?

Answer 7:

There are two different things here, and the report mixed them together, which is what invites this question.

The calibration step itself is doing a great deal of work. The fitted scale factors are 0.212 for dissolution, 0.317 for hardness, 0.570 for weight RSD and 0.255 for tensile. Because the raw predictive standard deviation is multiplied by that scale, the raw evidential intervals are between 1.75 and 4.72 times wider than they should be, depending on the target. Without the

correction, coverage would sit near 1.0 on every target, which is useless for deciding anything. So the step is very far from a no-op.

What barely moved is something much narrower: the difference between fitting that scale on the evaluation data and cross-fitting it on the other folds. Your reasoning about why is correct, and the explanation is that the scale factors are extremely stable across folds. For hardness they are 0.327, 0.311 and 0.312, varying by about 5 per cent. When the quantity being estimated is that stable, it makes very little difference which subset it is estimated from, so removing the leak changes almost nothing.

That stability is really the answer to your first question. A single multiplicative scalar per target, fitted on several hundred batches, is a very low-variance thing to estimate. It is not like fitting a model with many parameters, where a leak can inflate results badly. So the small change is what the situation predicts, not a warning sign.

The fix was still the right thing to do, because reporting a number fitted on its own evaluation data is not sound whether or not it changes the answer. But I should not have presented the small change as if it were reassuring or surprising. It follows directly from the scale being stable, and I should have explained it that way rather than leaving it as a bare observation.

To your last question directly. Two things would tell me the step was ineffective. First, scale factors close to 1, which would mean the raw intervals were already about the right width and the correction was doing nothing. Second, scale factors varying widely between folds, which would mean no single value was meaningful and the whole approach was unstable. Neither is the case here. The scales are far from 1 and they are consistent across folds, which is the combination that says the step is both necessary and reliably estimated.

Question 8

You use Kendall's tau to compare SHAP rankings between the most confident and least confident batches (Section 3.7 and 6.4), treating p less than 0.05 as evidence that the model shifts which sensors it relies on. But this comparison involves ranking only ten channels. How much statistical power does Kendall's tau really have at that scale, and could a similarly significant tau appear by chance if you randomly permuted which batches fell into the confident and unconfident groups?

Answer 8:

I will answer the power question first, and then raise something more important that this question led me to find.

On power. The exact two-sided p -value for $\tau = 0.822$ at $n = 10$ is 0.000358, which is what the report rounds to 0.0004. I confirmed this with the exact method in `scipy` and with a permutation check over 200,000 random ranking pairs, which puts the one-sided probability at roughly 0.0002. For contrast, $\tau = 0.467$ has an exact p of 0.0726, which is why the weight RSD and tensile result falls outside significance. So the p -values printed in the report are arithmetically correct for the test that was run.

On your permutation question, which is a different test, and which I did not run. I want to be clear about what I tested and what I did not.

What was tested: the permutation of the two *rankings*, described above. That is the check behind the 0.0002 figure.

What was not tested: the permutation you describe, of *which batches* land in the confident and unconfident groups. Those are different null hypotheses, and yours is the harder test, because it asks whether splitting by uncertainty does anything that a random split of the same batches

would not.

I did not perform that test, and I cannot perform it now. Notebook 11 saved the aggregated per-channel attribution ratios rather than the per-batch SHAP values, so the inputs needed to re-split the batches and recompute the rankings no longer exist outside a full re-run of the analysis. That is a limitation of how I saved my results. I would rather say that than give you the test I can run and let it stand in for the one you asked about.

I have no result to report for it, and I am not going to imply one. What I can give you is a prediction, and it is reasoning rather than evidence. Two random halves of the same batches would both be estimating the same underlying attribution, so their rankings should agree closely and τ should sit near 1. If that reasoning holds, a random split would give a *higher* τ than the 0.822 observed for the confidence-based split. Whether it does hold is unknown to me, because the measurement was never made. Re-running notebook 11 with the per-batch SHAP values retained, then permuting the group labels a few thousand times, is the experiment that would settle it, and it is the first thing I would run given more time.

Now the more important point. While checking this I found that the report interprets τ backwards, and I would rather raise it myself than leave it.

Section 6.4 states: “The rankings shift significantly on dissolution and hardness ($\tau = 0.822$, $p = 0.0004$).” But Kendall’s τ measures whether two rankings *agree*. A τ of 0.822 means they are strongly concordant, only four discordant pairs apart out of forty-five. The p -value tests the null hypothesis that $\tau = 0$, meaning no relationship at all. So $p = 0.0004$ says the confident and unconfident rankings are significantly *correlated*, which is evidence that the sensor ordering is **stable**, not that it changed.

The statistic I cited does not test the claim I made with it. That is an error in the report and I accept it.

The underlying finding is still real, but it is carried by different numbers. What actually changes when the model becomes uncertain about hardness is the magnitude of its reliance on each channel:

Channel	Ratio	Channel	Ratio
die fill	4.86×	tbl speed	1.38×
cyl main	2.81×	SREL	1.28×
main comp	1.58×	pre comp	1.15×
cyl pre	1.52×	ejection	1.01×
fom	1.46×	stiffness	0.98×

Read together with $\tau = 0.822$, the correct interpretation is that **the model does not reorder which sensors it uses. It keeps roughly the same ordering and leans far harder on two specific channels, die fill and main compression, when it is uncertain.** Ejection and stiffness barely move at all.

Before I say anything about what those ratios mean, I have to be straight about their status. They are descriptive numbers. I did not attach a significance test to them, and there is no confidence interval or standard error stored alongside them. The only inferential statistic anywhere in this analysis is τ , and τ tests rank agreement rather than the size of a ratio. So the amplification pattern is an observation from the data, not a statistically established result, and the report should not have called this the strongest statistical result in the study. That label came from τ ’s p -value, which I now accept was attached to the wrong claim.

One piece of supporting context that the results file does contain, and that I did not use in the

report. The mean epistemic uncertainty in the two groups is 6.688 for the confident fifth and 54.654 for the unconfident fifth, a separation of about 8.2 times. That is not a significance test either, but it does show the two groups are genuinely far apart on the variable used to split them, so the comparison is not being drawn between two groups that are effectively the same.

With that said, I would argue this is a more interesting finding than the one I claimed, because it is more specific. The model is not casting around differently when unsure. It concentrates on the two signals that physically govern how much material enters the die and how hard it is compressed, which is what determines tablet hardness. It was given ten raw channels and four laboratory numbers and was never told which of them mattered.

The wording in Section 6.4 should have described amplification of specific channels, evidenced by the ratios, rather than reordering, evidenced by τ .

Question 9

The report states plainly that this study covers one production line and one press (Section 6.5). Yet the framing throughout, in the title, the abstract, and the stated contribution, is pitched at the level of trustworthy deep learning for pharmaceutical manufacturing in general. If a second site or press were tested tomorrow, which parts of MT-TrajNet's design do you expect would fail to transfer, and why was the framing not scoped more narrowly given this limitation was already known?

Answer 9:

I will take the framing part first, because I do not really have a defence for it. The limitation was known from the start of the study, it is stated plainly in Section 6.5, and the title and abstract should have carried the same scope. "On industrial compression trajectories" reads as a general claim when the evidence is one press and one dataset. A more honest title would have named the dataset or described the work as a case study. I would change it if I could.

On what would fail to transfer, I would separate three layers.

What I would expect to transfer is the method rather than the model. Grouping cross-validation by product code to prevent leakage, cross-fitting the calibration scale so no batch is calibrated on its own prediction, testing distribution shift against a matched in-regime control, and stratifying attribution by the model's own confidence. None of these depend on the equipment. They are protocol, and they would apply to any instrumented press.

What would need refitting but not redesigning: the normalisation statistics, which are fold-specific in any case; the calibration scale factors, which would certainly differ on another press; and the 6,000-step sequence length, which was chosen for this dataset's run lengths and would need revisiting for a press with different batch durations.

What I would expect to fail is two things specifically.

The first is the learned encoder weights. The ten channels are specific to this press. Another press may carry different sensors, sample at a different rate, or measure the same nominal quantity differently. Channel ordering matters directly to my attribution result, so a different sensor set would invalidate the die fill and main compression finding until it was re-established from scratch.

The second, and the one I expect would fail hardest, is anything that depends on the hardness distribution. The $r = 0.941$ relationship says accuracy depends on how close a product sits to the centre of the training distribution. A press making a different product mix has a different centre, so the model would be accurate in a different place. Transfer would not simply be a

matter of the model performing worse; it would be a matter of it being good over a different range.

That last point points to what a useful transfer test would actually be. It would not be whether accuracy holds up, which I would expect to shift. It would be whether the calibrated uncertainty still flags unfamiliar regimes on new equipment. That is the property the thesis actually claims, and it is the one worth testing first.

Question 10

Your multi task model shares one encoder across four targets and trains with an unweighted sum of four evidential losses (Section 4.3). This is justified only by the targets being standardised to the same scale. But equal loss scale does not guarantee equal gradient contribution, especially once the evidential regularizer and the differing difficulty of each task come into play. How do you know that one of the four tasks, dissolution for instance, which shows the only significant multi task gain, is not simply dominating the shared representation rather than genuine positive transfer happening across all four targets?

Answer 10:

You are right that equal loss scale does not guarantee equal gradient contribution, and I did not monitor per-task gradient norms during training. So I cannot rule out an imbalance by direct measurement. That is a genuine gap in the study.

What I can do is look at the outcome, because the domination idea makes a prediction. If dissolution were monopolising the shared encoder, the other three targets should be damaged by sharing, or at best left unaffected.

Target	Multi-task	Single-task	Improvement	<i>p</i>
Dissolution	3.261	3.547	8.1%	0.012
Hardness	7.698	8.310	7.4%	0.800
Tensile	0.345	0.364	5.2%	0.830
Weight RSD	0.585	0.595	1.7%	0.904

On the fold means, all four targets improve, and hardness gains 7.4 per cent, almost as much as dissolution's 8.1 per cent.

But I should not leave it there, because the fold means hide something. Looking at the per-fold differences rather than the averages, the picture is less uniform than that table suggests:

Target	Per-fold difference (single – multi)	Multi-task better on
Dissolution	+0.310, +0.246, +0.301	3 of 3 folds
Hardness	+3.154, +0.086, –1.404	2 of 3 folds
Weight RSD	–0.002, +0.098, –0.066	1 of 3 folds
Tensile	–0.019, –0.039, +0.114	1 of 3 folds

Only dissolution improves on every fold. Weight RSD and tensile are worse under sharing on two folds each, and their mean-level gain comes from one fold in both cases. Hardness improves on two folds and is worse on the third.

So the honest version of my argument is narrower than the one the report implies. What the evidence does show is that dissolution, the task you suggest might be dominating, is not gaining at the expense of hardness: hardness improves on two folds, including a large gain of 3.154 on fold 1. That is not what domination by dissolution would look like. What the evidence does not show is consistent positive transfer to all four targets, and I should not have implied it does.

Dissolution reaches significance because its per-fold differences are consistent, all three between +0.246 and +0.310. The other three do not, because their differences change sign between folds, and a paired test on three folds cannot find an effect when the sign flips.

I want to correct something I could easily have got wrong here. It is tempting to blame this on MT-TrajNet being unstable across folds, but that is not what the data says. The multi-task hardness runs are stable, at 7.900, 7.428 and 7.767, a spread of 1.06 times. It is the single-task hardness runs that swing, at 11.054, 7.514 and 6.363, a spread of 1.74 times. The instability is in the baseline I am comparing against, not in my own model.

There is also a physical reason to expect genuine transfer rather than one task taking over. Hardness and tensile strength both depend on compaction, and dissolution depends on it as well. The four targets share upstream causes in the process, which is exactly the setting where a shared representation should help several tasks at once rather than only one.

So my position is that what I observed does not fit domination, because domination would mean the other tasks got worse, and all four got better on the means. But I accept this is indirect evidence about the effect, not direct evidence about the gradients themselves. The direct tests would be logging per-task gradient norms through the shared encoder during training, or comparing against a learned task-weighting scheme. I ran neither.

There is one more thing I should put on the record here, because it goes to the heart of what you are asking and I would rather raise it than have it found. My proposal specified the uncertainty weighting scheme of Kendall et al. (2018), which treats each task's homoscedastic uncertainty as a learnable parameter and balances the tasks automatically. That is exactly the mechanism that would have answered your concern, because it removes the assumption that equal loss scale means equal gradient contribution. The model I built does not use it. It uses an unweighted sum instead.

I have gone back through the submitted report and it does not flag that change anywhere. Section 4.3 presents the equal sum as the design choice and argues that standardising the targets makes it sufficient, without noting that something different had been planned. That is a gap in the report. The change should have been stated and justified rather than passed over quietly, and I never tested whether the simpler choice cost anything.

Closing note

Several of these questions found things I had not checked, and two of them found genuine errors in the submitted report: the missing multiple-comparison correction raised in Question 1, and the misinterpretation of Kendall's τ raised in Question 8. I would rather set both out plainly than argue around them.

Where that leaves the study is worth setting out plainly. The out-of-distribution result is the part I am most confident about, because it was attacked from two directions and held: the uncertainty tracks a real change of operating regime rather than mere novelty, and that survives both the 24-code analysis and the within-code-13 check. The attribution finding is real as an observation, but it is about amplification of two specific channels rather than reordering, and it has no significance test behind it. The multi-task result is narrower than the report suggests.

Dissolution improves on every fold, while weight uniformity and tensile strength are worse under sharing on two folds each, so I cannot claim consistent transfer across all four targets.

The weaknesses reported in the thesis remain weaknesses, and Question 3 has shown me there was a further one I should have investigated.